Application layer

1. Short notes on —

(i) **DNS :** DNS stands for Domain Name System, which is a hierarchical naming scheme used in the internet.

1. DNS is a mapping system for the web address to their IP address
2. When a web address is typed in the browser it actually sends a broadcast message to its network, who knows the IP address of corresponding web address.
3. The DNS server replies back the answer with IP address.

The DNS is divided into over 200 - top level domains. These top level domain comes in two types :

i) generic such as com (commercial), edu (educational)
ii) Countries.

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(ii) HTTP : The transfer Protocol used throughout the WWW is HTTP. HTTP stands for HyperText Transfer Protocol. It specifices what message, client may send to servers and what response they get back in return.

(iii) SMTP (Simple Mail Transfer Protocol) :

SMTP is a simple ASCII Protocol. After establishing connection to Port 25 the sender operated as a client and the reciever operated as server. The Server starts by sending a line of text giving its identity & telling whether it is Prepared to recieve. If not the client releases to accept then the client ~~releases to accept~~ announces source & destination address. If such recipient exists message is forwarded to the reciever.

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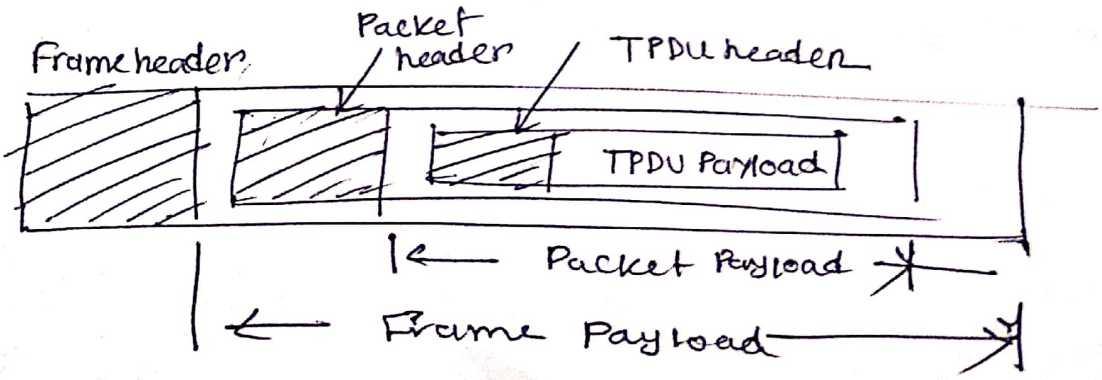
(iv) **DHCP :** In networks, with a large number of hosts, Statically assigning IP address and other IP information quickly become impractical.

Dynamic Host Control protocol (DHCP) Provides administrators with a mechanism to dynamically allocate IP addresses, rather than manually setting the addresses on each device.

DHCP Can Provide the following Protocol Parameters —

1. Default gateway
2. Domain name & DNS servers
3. Time servers
4. WINS servers

(v) **TPDU :** The transport Protocol Data unit is the form of the message at transport layer. It contains the original message and a TPDU header which basically contains the Port number of the relevent data. The original message is kept in the TPDU Payload field.



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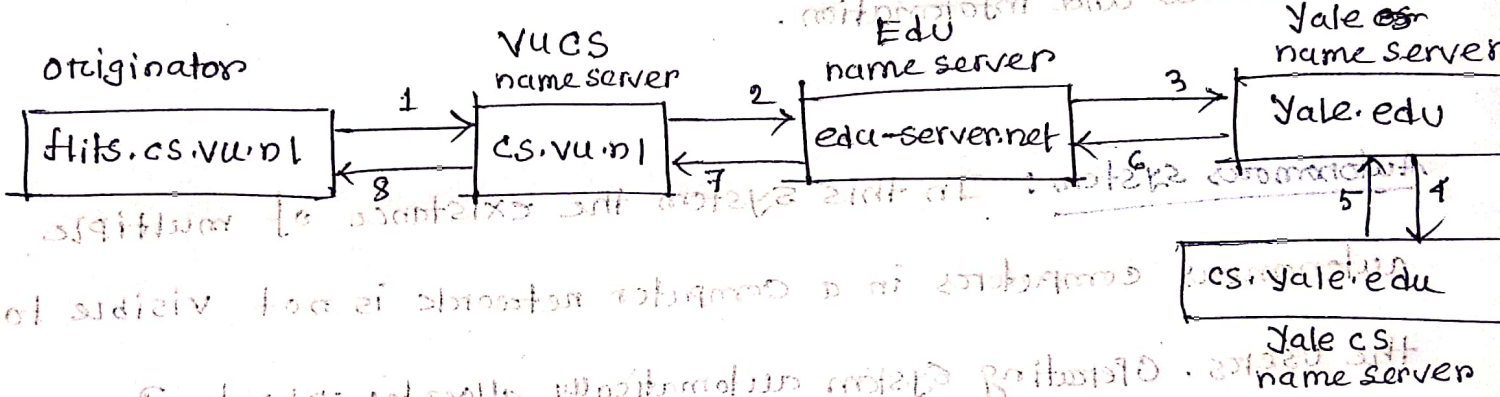
2. How a resolver looks up a remote name in eight steps?

To map a name onto an IP address an application program calls a library procedure called the resolver, passing it the name as a parameter. In the following figure, a resolver on flits.cs.vu.nl wants to know the IP address of the host linda.cs.yale.edu. In step 1, it sends a query to the local server, cs.vu.nl.



Let's suppose the local name server has never had a query for this domain before and knows nothing about it. It may ask a few other nearby name servers, but if none of them know, it sends a UDP packet to the server for edu given in its database edu-server.net. It is unlikely that this server knows the address of linda.cs.yale.edu & cs.yale.edu, but it must know all of its own children so it forwards the request to the name server for yale.edu (step 3). In turn, this one forwards the request to cs.yale.edu (step 4). Since each request is from a client to server, the resource record requested works its way back in step 5 through 8.

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